Time Series Modelling Case Study

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Github Link: https://github.com/dekshinaudayan/german-electricity-demand-forecasting

Abstract

Forecasting electricity consumption plays an important role in the efficient functioning of power systems in terms of generation scheduling, electricity trading, etc. The current paper aims to analyze statistical, machine learning, and deep learning models to forecast German electricity demand with the use of OPSD dataset for the period of January 2015 – September 2020. Firstly, benchmark models such as Mean, Naive, Drift, and Seasonal Naive were constructed in order to provide the baseline for further research. Then, more sophisticated models such as SARIMA, SARIMAX, XGBoost, and hourly LSTM network were built. Temperature data from Berlin were included into SARIMAX and feature-based models as predictor variable. Forecast accuracy was measured by means of RMSE, MAE, and MAPE metrics. It has been found out that sophisticated machine learning models were more efficient than benchmark and statistical ones and hourly LSTM model produced the best predictions among all analyzed models.

1. Introduction

Proper electricity demand forecasting helps maintain the quality and efficiency of electricity supply services. Forecasting helps with planning of generation scheduling, system operations, and markets; at the same time, operational costs are reduced. Electricity demand is affected by seasonality, weather conditions, and behavior of consumers, which makes it a difficult forecasting task. This research evaluates statistical, machine learning, and deep learning methods to forecast German electricity demand. Hourly electricity demand time series for January 2015 - September 2020 were analyzed from the data provided by the Open Power System Data (OPSD) project. First, exploratory data analysis and stationarity tests were conducted to describe time series properties. Next, benchmark models and advanced models including SARIMA, SARIMAX, XGBoost, and hourly LSTM neural network models were created and compared. In addition, temperature information from Berlin was included as explanatory variables for SARIMAX and feature-based models. The performance of models was assessed using RMSE, MAE, and MAPE metrics. The goal of the research is to find the most suitable forecasting method that will give good forecasting accuracy and will be interpretable and applicable in practice.

2. Dataset Description

The time series of electricity demand in Germany was sourced from the Open Power System Data (OPSD). The OPSD dataset comprises of electricity load data collected at hourly intervals from January 2015 to September 2020. The hourly time series was converted to daily and weekly averages using Pandas to facilitate exploratory and statistical forecasting tasks. Historical weather temperature data for Berlin were sourced from the Open-Meteo Archive API and converted to weekly averages. In addition, Heating Degree Days (HDD) and Cooling Degree Days (CDD) were computed and combined with the weekly average electricity demand data.

3. Exploratory Data Analysis

The hourly, daily, and weekly demand for electricity was graphed in order to observe the overall trends. The demand for electricity clearly had an annual seasonal pattern whereby electricity demand was high during winter but low during summer.

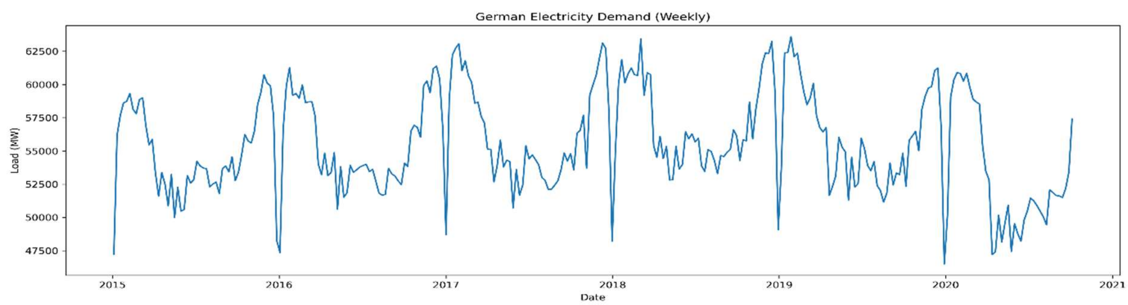

Figure 1: Weekly Electricity Demand

A seasonal decomposition of the weekly data was then carried out to determine the different components of the time series. It was found that there was a very clear seasonal trend each year as well as a slow changing long-term trend. This implies that seasonality forecasting models can be used in modeling the data.

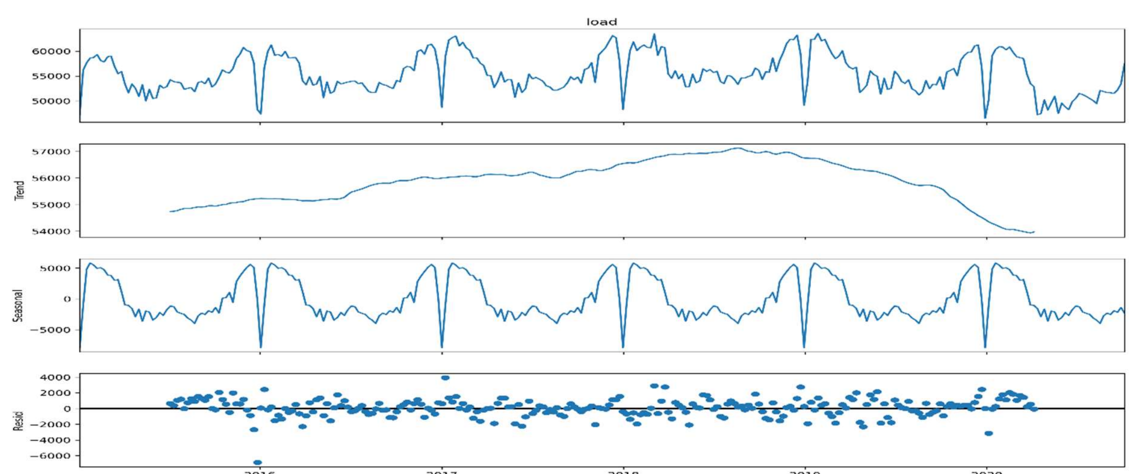

Figure 2: Seasonal Decomposition of Weekly Electricity Demand

These patterns fit into the predicted behavior of electricity consumption that increases with lower winter temperatures due to heating needs and decreased industrial operations during holidays. Based on these features, it seems appropriate to use seasonality and weather factors for forecasting purposes.

4. Stationarity Analysis

The stationarity was checked by visual check, seasonal decomposition, and statistical hypothesis testing methods. According to the result of the seasonal decomposition method (figure 2), there was a strong seasonal pattern and trend over one year which showed that the initial time series is not stationary. To eliminate seasonality, the method of seasonal differencing with a lag of 52 was used. The null hypothesis of the ADF test was rejected while KPSS test shows that the difference series is stationary. Thus, the decision about applying the order of seasonality D=1 and order of non-seasonal differencing d=1 could be made. Also, annual seasonality justified the use of seasonal period of 52 weeks for both SARIMA and SARIMAX models.

Table 1. Stationarity test results

Test	Statistic	P-Value	Conclusion
ADF	-4.047	0.0012	Stationary after differencing
KPSS	0.160	0.1	Stationary after differencing

5. Benchmark Forecasting Models

To establish baseline performance, four benchmark forecasting models were evaluated using the final two years of weekly electricity demand as the test set:

- Mean Forecast
- Naive Forecast
- Drift Forecast
- Seasonal Naive Forecast

The Seasonal Naive model performed better than other benchmark models due to the presence of the powerful annual seasonality evident in the dataset. It thus forms the basis against which other forecasting techniques to be explored in the subsequent sections are compared.

6. SARIMA Model

The SARIMA model was built to incorporate the trend and annual seasonality exhibited by the weekly electricity demand series. A thorough grid search was conducted on all possible combinations of autoregressive (p = 0-6), differencing (d = 0-2), and moving average (q = 0-6) terms based on the Akaike Information Criterion (AIC). The seasonal terms were set to

(0,1,1,52) to model the annual seasonality with a period of 52 weeks. The SARIMA(2,2,6)(0,1,1,52) model with the minimum AIC was chosen as the final forecast model. Forecasts generated from the model were close to the actual weekly electricity demand series, incorporating the seasonality and smoothing out the short-term variations. From residual analysis, it was found that there was no autocorrelation in the residuals of the model, implying that the model incorporated the entire information in the data series.This demonstrates that selecting a model based solely on AIC does not always result in the best out-of-sample forecasting performance.

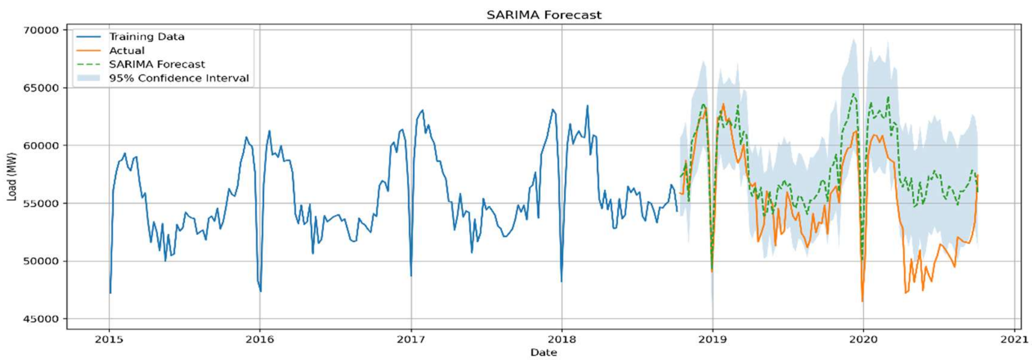

Figure 3: SARIMA Forecast with 95% confidence intervals.

7. SARIMAX Model

In order to examine the influence of weather on electricity demand, weekly temperature information was added to the SARIMA model to develop a SARIMAX model. The weekly mean temperature along with HDDs and CDDs were included in the model as exogenous variables. As observed future temperatures were considered for the testing purpose, the model can be called an explanatory or conditional forecast.

Although incorporating temperature data made the model more capable of capturing weather-driven changes in demand pattern, there wasn't much improvement in the forecasting accuracy. Furthermore, the coefficient estimates suggested that individual temperature variables were not significant at all, implying that annual seasonality had captured most of the demand variations.

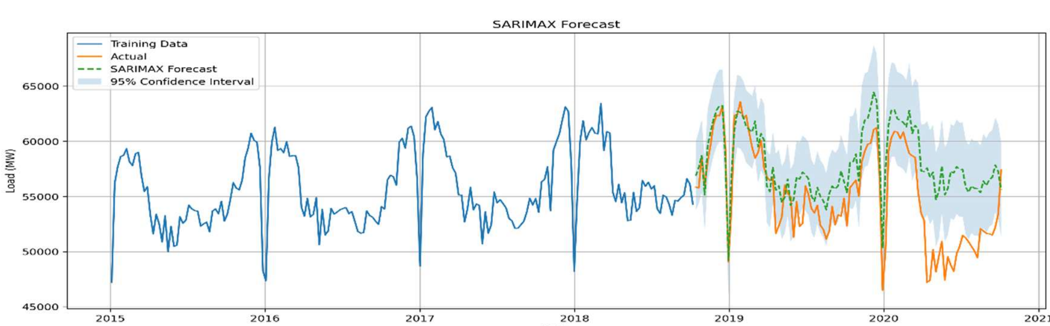

Figure 4. SARIMAX forecast using temperature variables.

8. Feature-Based Regression Model

The XGBoost model for regression was built with the use of calendar-based features as well as weekly temperatures. The features used during engineering were cyclical month and week features, quarterly features and weather features. Using these features allowed the algorithm to detect nonlinear dependencies which could not be captured by statistical models.

The XGBoost model yielded considerably lower forecast errors compared to the benchmarks such as SARIMA and SARIMAX models, proving the efficiency of using feature-based machine learning for forecasting electricity demand. It is important to note that seasonality-related calendar features had the highest impact on the forecast. The XGBoost model produced substantially lower forecasting errors than the statistical models. Performance metrics are summarised in **Table 2**.

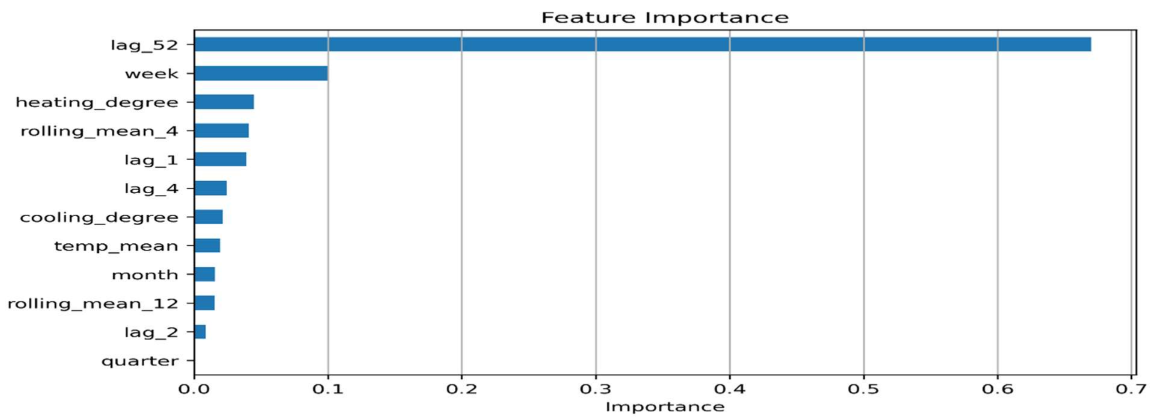

Figure 5. Feature importance obtained from the XGBoost model.

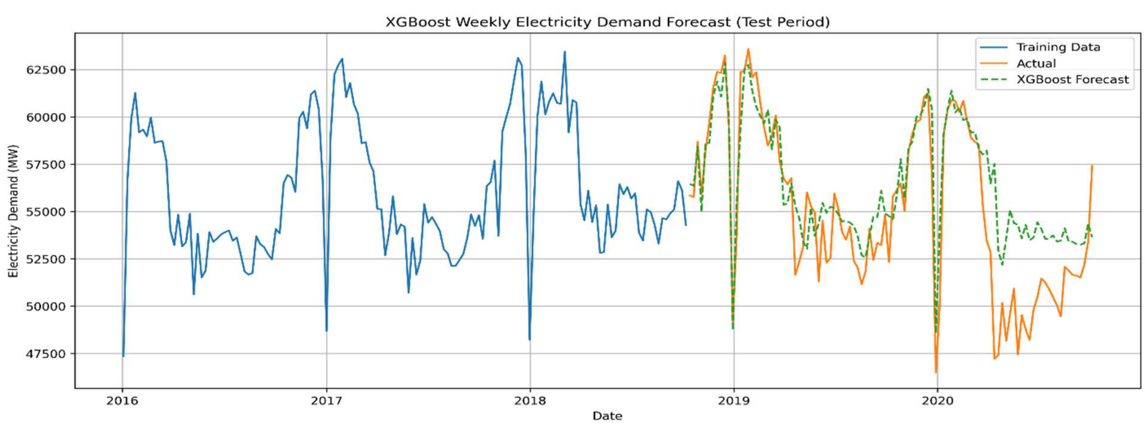

Figure 6: XGBoost weekly electricity demand forecast for the test period

9. Long Short-Term Memory (LSTM) Model

A Long Short-Term Memory (LSTM) neural network model was constructed for predicting hourly electricity consumption demand. In contrast to the previously presented statistical and machine learning methods created using weekly electricity consumption data, this model was based on the raw time series of hourly electricity consumption demand that allows for taking into account not only short-term dependencies but also nonlinear associations.

Firstly, all data were normalized using Min-Max normalization method, and then, the input sequence of the last 24 hours was used for predicting the next hour of electricity consumption demand. The initial dataset was split into a training set and 17,520-hour (about two years) test period. Finally, a model consisting of one LSTM layer and 64 hidden units and one dense layer was trained using Adam optimization method and mean square error criterion as a loss function.

The LSTM model outperformed other forecasting methods. As it can be seen from Figure 7, the obtained forecasts of electricity consumption demand matched perfectly well the real values during the test period, correctly predicting daily and seasonality effects. A comparison of evaluation metrics for all models is presented in **Table 3**.

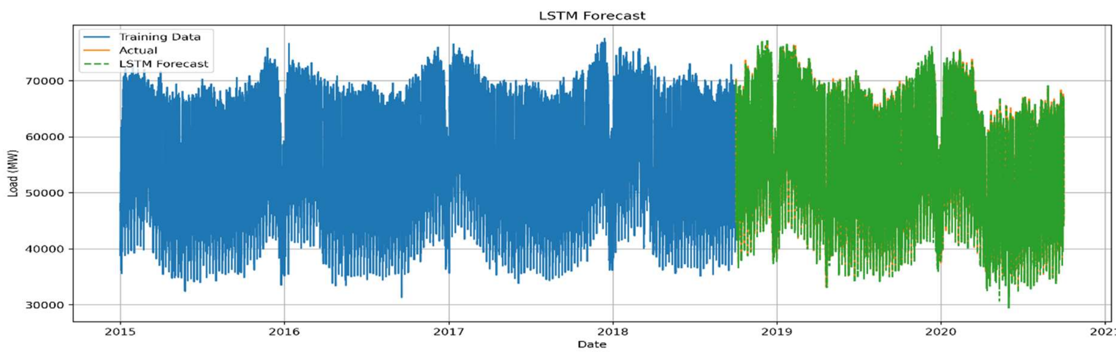

Figure 7. LSTM hourly electricity demand forecast for the test period.

Table 2. Comparison of forecasting performance

Model	**Data**	**RMSE**	**MAE**	**MAPE**
Mean	Weekly	4397.29	3788.83	6.969
Naive	Weekly	4459.11	3783.20	6.79
Drift	Weekly	5117.95	4339.89	8.049
Seasonal naive	Weekly	3006.76	2318.52	4.409
SARIMA	Weekly	4872.99	3991.13	7.563
SARIMAX	Weekly	3851.14	3069.89	5.858
XGBoost	Weekly	2509.13	1787.89	3.436
LSTM	Hourly	643.18	490.11	0.912

Seasonal Naive was an effective base model due to the significant seasonality present in the data. While the SARIMA and SARIMAX models managed to incorporate seasonality in their models, they failed to outperform the Seasonal Naive base model during testing. XGBoost reduced forecast error significantly through modeling non-linear relationships, but the hourly LSTM model had the most accurate forecasts.

10. Discussion

1. Comparison with the Seasonal Naive Benchmark

Seasonal naive performed very well because the electricity consumption showed seasonality every year on a weekly basis. On the other hand, SARIMA, SARIMAX, XGBoost, and LSTM outperformed the Seasonal Naive with fewer forecasting errors. Out of these models, the LSTM generated the best result.

2. Avoiding Data Leakage

Data leakage was prevented through the generation of temperature features using data available for each particular training and testing period. While the forecasting models were trained on historical data, the test period was completely hidden from them. In the case of SARIMAX, actual temperature values for the future were considered only explanatory variables when evaluating the forecast.

3. SARIMAX Differencing and Seasonal Period

The stationarity test showed that the original weekly data had both a trend component and a seasonal component with a cycle of one year. This means that the difference orders used for the time series data were d=1 and D=1. The value of the seasonal period was 52 weeks.

4. Effect of Temperature Covariates

Incorporating temperature variables into the model helped capture weather variations on electricity demand. Yet the forecast performance gains were rather small, implying that most of the seasonal differences had already been considered in the SARIMA part. Since future values of temperatures are unknown in reality, the temperature variables need to be substituted for weather forecasts.

5. Model Complexity and Interpretability

The advantage of SARIMAX is that each parameter is associated with certain characteristics like autoregression, moving average, and seasonal factors. XGBoost is flexible in prediction, but still somewhat interpretable due to the feature importance analysis. Meanwhile, LSTM proved to be the most accurate one, but it works as a black box.

6. Recommended Model

It is suggested to use LSTM model for forecasting operation electricity demand since it produced the lowest forecasting error and was able to correctly detect the complex pattern within the hourly electricity demand series. Nevertheless, in case interpretability and simplicity of maintenance are important, there is an alternative model - XGBoost.

11. Conclusion

This research compared various benchmark statistical techniques, SARIMA, SARIMAX, XGBoost, and LSTM models in order to forecast German electricity consumption. Initial data analysis showed the existence of strong yearly seasonality, which became the reason for applying seasonal forecasting models. It was found that modern machine learning and deep learning models performed much better than benchmark models. In particular, hourly LSTM forecasts showed the best results, and XGBoost model was the optimal choice because of the good trade-off between forecasting quality and model transparency.

12. Future Work

Potential avenues for future research can be the development of forecasting models which will integrate both statistical and deep learning techniques. Another possible area is the inclusion of more explanatory variables such as public holidays, renewable energy production, and economic indicators. Other possibilities include hyperparameter optimisation and attention-based deep learning frameworks. Future work could also investigate probabilistic forecasting and transformer-based deep learning architectures.

13.References

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